

Assignment No: - 5

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Problem Statement

Use WordNet to identify semantic relationships such as synonyms , antonyms and hypernymy in text data.

Objectives

To use **NLTK WordNet** to extract:

- Synonyms
- Antonyms
- Hypernyms
- Hyponyms

S/W Packages and H/W apparatus used

- **Operating System:** Windows / Linux / macOS
- **Programming Language:** Python 3.x
- **Development Tools:** Jupyter Notebook / Google Colab / VS Code
- **Library Used:** NLTK (Natural Language Toolkit)

Hardware Apparatus Used

- Computer or Laptop
- Processor: Intel / AMD or equivalent
- Minimum **8 GB RAM**
- Internet connection (for downloading NLTK resources)

Theory

What is WordNet?

WordNet is a lexical database of English words where words are grouped into sets called synsets (synonym sets).

Each synset represents a specific meaning of a word.

It also stores semantic relationships like:

Relationship	Meaning
Synonym	Words with similar meaning
Antonym	Words with opposite meaning
Hypernym	A more general word (parent)
Hyponym	A more specific word (child)

♦ 2. What is a Synset?

A **synset** is a group of words that share the same meaning.

Example:

Happy → glad, joyful, cheerful

Explanation of the Program

Your code uses **NLTK** to access WordNet and find different semantic relations.

● A) Finding Synonyms of *Sunshine*

```
synsets = wn.synsets("Sunshine", pos=wn.ADJ)
```

- It searches for **adjective meanings** of “Sunshine”.
- Each synset contains similar words.
- Lemmas are extracted → stored in a set (no duplicates).

✓ Output may include words related to brightness or cheerfulness.

B) Finding Antonyms of *sad*

for syn in wn.synsets("sad", pos=wn.ADJ):

for lemma in syn.lemmas():

if lemma.antonyms():

antonyms.add(lemma.antonyms()[0].name())

- For each meaning of *sad*, WordNet checks opposite words.
- Example antonyms: **happy, glad, cheerful**

C) Finding Hypernyms of *dog*

for syn in wn.synsets("dog", pos=wn.NOUN):

for hypernym in syn.hypernyms():

Hypernym = General Category

Dog → animal → mammal → organism

So WordNet returns broader classes of *dog*.

D) Finding Hyponyms of *dog*

4 for hyponym in syn.hyponyms():

Hyponym = Specific Type

Dog → bulldog, poodle, terrier, shepherd dog, etc.

Concepts Used

Concept	Meaning in this Program
Corpus	WordNet dataset
Synset	Set of synonyms
Lemma	Base word form
Semantic Relation	Meaning connection between words
Set Data Structure	Removes duplicates

Conclusion

This experiment demonstrates how **NLTK WordNet** helps in understanding **semantic relationships** between words.

We successfully extracted:

- Synonyms (similar meaning words)
- Antonyms (opposite words)
- Hypernyms (general categories)
- Hyponyms (specific types)

This technique is widely used in:

- Chatbots
- Text classification
- Sentiment analysis
- Search engines
- AI language understanding

It shows how machines can understand **word meaning relationships**, which is a core part of **Natural Language Processing (NLP)**.
